

Ayanava Dasgupta

Curriculum Vitae

Contact Information

Correspondence Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata,
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Web Profiles Google Scholar | arXiv

Research Interests

Quantum Communication, Quantum Machine Learning, Classical and Quantum Information Theory,
Quantum Statistical Learning, Quantum Hypothesis Testing etc.

Academic Qualification

- 2023–Present **PhD in Computer Science**, *Indian Statistical Institute, Kolkata*, Advisor: **Dr. Naqueeb Ahmad Warsi**
- 2025–Present **Senior Research Fellow**, *Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata*
- 2023–2025 **Junior Research Fellow**, *Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata*
- 2021–2023 **M.Tech in Computer Science**, *Indian Statistical Institute, Kolkata*, 85.3%
- 2017–2021 **B.Tech in Computer Science and Engineering**, *Government College of Engineering and Leather Technology, Kolkata*, DGPA 9.51/10

Peer-Reviewed Publications

- April 2026 **Generalization Bounds for Quantum Learning via Rényi Divergences**
with Naqueeb Ahmad Warsi and Masahito Hayashi
IEEE Transactions on Information Theory | DOI: 10.1109/TIT.2026.3683678 | arXiv:2409.05619
- Feb 2025 **Universal tester for multiple independence testing and classical-quantum arbitrarily varying multiple access channel**
with Naqueeb Ahmad Warsi and Masahito Hayashi
IEEE Transactions on Information Theory | DOI: 10.1109/TIT.2025.3538870 | arXiv:2409.05619
- Jan 2025 **Intersection and union of subspaces with applications to communication over authenticated classical-quantum channels and composite hypothesis testing**
with Naqueeb Ahmad Warsi
IEEE Transactions on Information Theory | DOI: 10.1109/TIT.2025.3531565 | arXiv:2311.10524

Preprints and Working Papers

- Sep 2026 **Haar-Bayesian Pure-State Prediction under Relative-Entropy Loss: Arbitrary-Effect Reduction and Global Optimality**
with Masahito Hayashi and Naqueeb Ahmad Warsi | arXiv:2609.10072
- Sep 2026 **Uniqueness and Cramér-Rao Efficiency of Quantum U-Statistics**
with Naqueeb Ahmad Warsi and Premanshu Chatterjee | arXiv:2609.08745
- June 2026 **Privacy Implies Stability: Information-Theoretic Generalization Bounds for Quantum Learning**
with Naqueeb Ahmad Warsi and Masahito Hayashi | *Communicated at Quantum* | arXiv:2602.01177
- Nov 2025 **Quantum Information Ordering and Differential Privacy**

Fellowships and Achievements

- 2026 Qualified UGC National Eligibility Test (NET) for Assistant Professorship in Computer Science and Applications with 98.91 percentile.
- 2025 Recipient of TCS Research Fellowship (Cycle 19)
- 2021 98.8 Percentile in Graduate Aptitude Test in Engineering (GATE)

Research Talks

- April 2025 **Generalization Error Bounds of Quantum Learning Algorithms via Rényi divergences**
Presented at *Advanced Computing and Micro-electronics Unit (ACMU)*, Indian Statistical Institute, Kolkata

Teaching Experience

- Jul–Nov 2026 **Teaching Assistant**, *Abstract Algebra and Number Theory*, Indian Statistical Institute, Kolkata, Instructor: Dr. Sumanta Ghosh
- Jul–Nov 2025 **Teaching Assistant**, *Coding Theory*, Indian Statistical Institute, Kolkata, Instructor: Dr. Sumanta Ghosh

Thesis and Reports

- Master's Thesis **A Study on Union and Intersection of Subspaces and its Application to Quantum Communication** (June 2023)
Advisor: Dr. Naqeeb Ahmad Warsi | Indian Statistical Institute, Kolkata
Explored typical set operations translated to quantum information theory, building a novel notion of the intersection of subspaces for constructing POVMs.
- Minor Project **Expressibility and Entangling Capability in Noisy Parameterized Quantum Circuits** (July 2023)
Advisor: Dr. Susmita Sur-Kolay | Indian Statistical Institute, Kolkata
Examined the behavior of Expressibility and Entangling Capability metrics under noise consideration by investigating 19 Parameterized Quantum Circuits across varying complexities.

References

- Dr. Naqeeb Ahmad Warsi** (PhD Advisor)
Associate Professor, Computer and Communication Sciences Division,
Indian Statistical Institute, Kolkata
Email: nauqueebwarsi@isical.ac.in
- Prof. Masahito Hayashi** (Collaborator)
Professor, School of Data Science
The Chinese University of Hong Kong, Shenzhen
Email: hmasahito@cuhk.edu.cn